

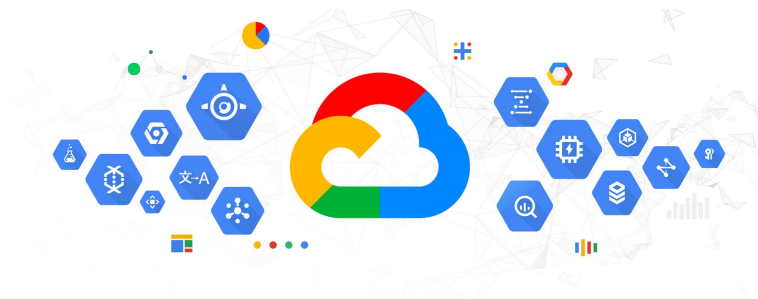
Deploying Your Java REST Application to Google Cloud

A Step-by-Step Guide

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What is Google Cloud Platform (GCP)?

- **Google Cloud Platform** is a suite of cloud computing services offered by Google that provides a series of **modular cloud services** including computing, data storage, data analytics, and machine learning, alongside a set of management tools.
- Developers can **build, deploy, and scale** applications and websites using Google's powerful infrastructure.
- Other similar cloud platforms: Amazon Web Services (AWS), Microsoft Azure

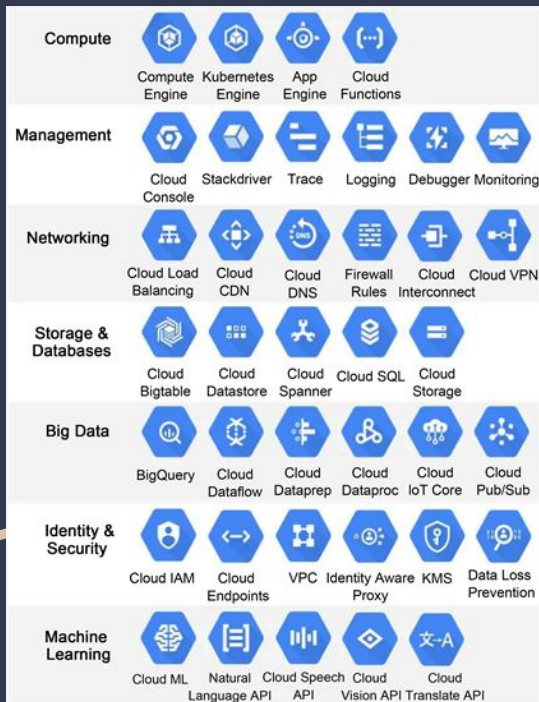


Why GCP?

- **Scalability:** You can easily add or remove resources as needed.
- **Cost-effective:** You only pay for what you use.
- **Reliability:** Google's infrastructure is highly reliable and secure.
- **Global reach:** Your applications can be accessed from anywhere in the world.



Some popular GCP services



Compute

- **Compute Engine:** Virtual machines for running applications.
- **App Engine:** Platform for building and deploying web applications.

Storage

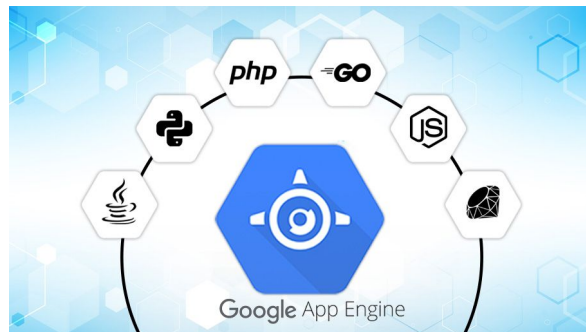
- **Cloud Storage:** Object storage for storing files and data.
- **Cloud SQL:** Managed relational database service.
- **Cloud Firestore:** NoSQL database for scalable applications.

Database

- **Cloud Bigtable:** NoSQL database for large-scale, high-performance analytics.
- **Cloud Spanner:** Globally distributed, strongly consistent relational database.
- **Cloud Dataflow:** Managed service for data processing and analysis.

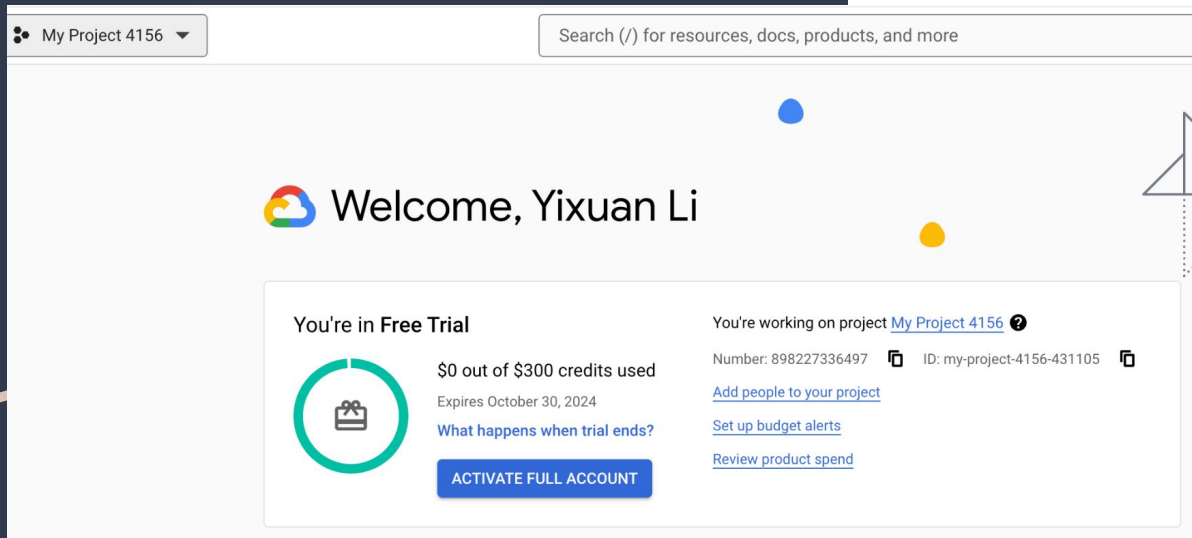
Google App Engine

- Key feature: a **fully managed platform** that allows developers to build, deploy, and scale web applications. It handles much of the underlying infrastructure, making it easier for developers to focus on writing code.
- Ideal for: serverless applications, auto-scaling, managed environment
- Supported languages: Go, Java, Node.js, PHP, Python, Ruby
- <https://cloud.google.com/appengine/docs/an-overview-of-app-engine>



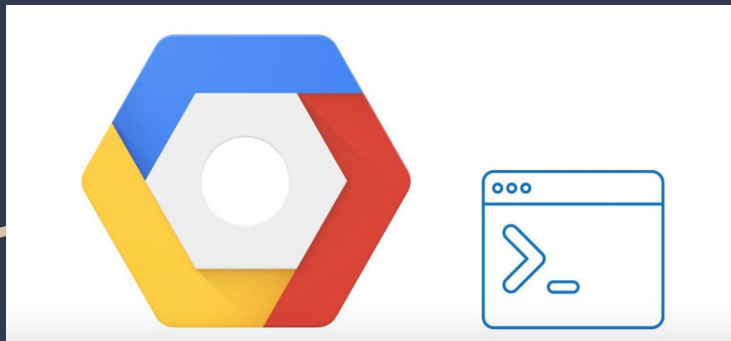
1. Creating your project

- Go to the Google Cloud Console.
- Click on the project drop down in the top left corner and select "New Project".
- Enter a name and billing information for your project, then click "Create".
- <https://cloud.google.com/resource-manager/docs/creating-managing-projects>



- **Project name:** A human-readable name for your project. The project name isn't used by any Google APIs.
- **Project ID:** A globally unique identifier for your project.
- **Project number:** An automatically generated unique identifier for your project.

2. Install and initialize the gcloud CLI



- Download and Install:
<https://cloud.google.com/sdk/docs/install>
- Alternatively, if you are on mac, you can install it via Homebrew:
<https://jansutris10.medium.com/install-google-cloud-sdk-using-homebrew-on-mac-2952c9c7b5a1>
- Authenticate Google Cloud Developer account:
`gcloud auth login`
- Initialize the SDK: Run `gcloud init` and follow the prompts to authenticate and set up your SDK.
<https://cloud.google.com/sdk/docs/install-sdk>

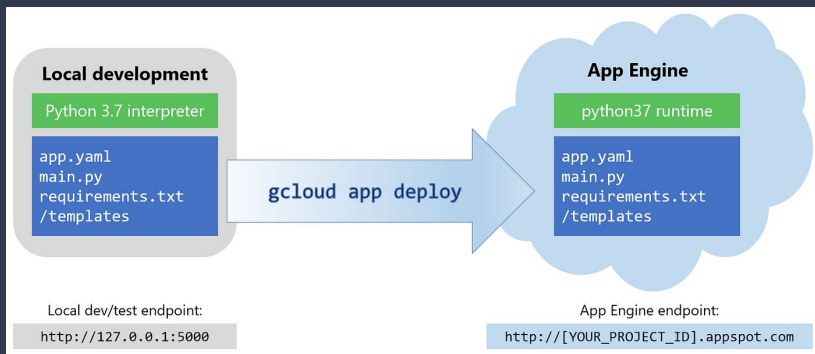
3. Creating the app.yaml file

- It defines the application's environment, deployment settings, and resource allocation. This file is used to specify how your application should be deployed, scaled, and managed on the App Engine platform.
- Add an `app.yaml` File (under `IndividualProject`)
- App Engine `app.yaml` reference:
<https://cloud.google.com/appengine/docs/standard/reference/app-yaml?tab=java>

```
runtime: java21 # or another supported runtime
entrypoint: java -jar YOUR-ARTIFACT.jar
instance_class: F2

env_variables:
  BUCKET_NAME: "example-gcs-bucket"
```


4. Deploy your web service



- If you haven't specify the project during `gcloud init`, run `gcloud config set project PROJECT_ID`.
- To deploy your service: `gcloud app deploy`
- Open your deployed Google App Engine application in a web browser: `gcloud app browse`
 - You could see the URL of your app in the terminal and the browser.
- Test your app endpoint
 - `curl --request GET --url ${YOUR_APP_URL}`
 - Postman: <https://www.postman.com/>

```
~/IdeaProjects/4156-Miniproject-2024-Students/IndividualProject y13803*  
> curl --request GET --url [REDACTED]  
Welcome, in order to make an API call direct your browser or Postman to an endpoint  
  
This can be done using the following format:  
▶ http://127.0.0.1:8080/endpoint?arg=value
```

5. Manage and update your application

Use the App Engine dashboard in the GCP Console to monitor and manage your app.

Monitoring and Updates

- **App Engine logs:** Use the Google Cloud Console or the `gcloud app logs` command to view application logs and troubleshoot issues.
- **Versioning:** Create multiple versions of your application to test changes before deploying them to production.
- **Deployment:** Use the `gcloud app deploy` command or the Google Cloud Console to deploy new versions of your application.



App Engine





Dashboard



Services



Versions

Services

 DELETE

 EDIT INGRESS SETTING

☒	Service	Versions	Labels	Dispatch routes	Ingress 	VPC access name 	VPC egress setting 	Last version deployed 	Diagnose
☒	default 	2			All			Jul 31, 2024, 3:08:29 AM by yixli9811@gmail.com	Logs 

Resources

- <https://cloud.google.com/appengine/docs/standard/java-gen2/building-app>
- <https://cloud.google.com/appengine/docs/standard/testing-and-deploying-your-app?tab=java>
- Other approaches to deploy your application to GCP
 - Cloud Run: container-based
 - Compute Engine: virtual machines